Michael Remington Coulter

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PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer Apr 2023 - Present

Interconnect Architect | CMN GUI Lead | Tools for internal engineers and external customers.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025), for a project-agnostic multi-agent orchestration and context-gathering system.

- Built a Rust/Tauri 2 workbench with React/TypeScript UI, a shared desktop/CLI core, bounded asynchronous Git inspection and verified process controls.
- Built isolated workspaces with dedicated ports, caches and editor state; coordinated concurrent checkout allocation through atomic locks, tokenized leases and ownership-checked recovery.
- Built a Python NIC-400 XML/CCXML migration engine that reconstructs topology and routes while preserving placement; historical validation recorded 4,130/4,130 valid configurations passing design-rule checks and 5/5 curated invalid configurations rejected.
- Created the NCI/NoC logical canvas and console for configuration logs and design-rule results; built canvas exports and configuration-report scripts.
- Introduced Pixi.js to improve NoC canvas performance and created Playwright test infrastructure for newer non-coherent interconnect tooling.
- Prevented stale or concurrent design-rule responses from updating the wrong view using request IDs and reload checks; added unit and end-to-end regressions.

Software Engineer Apr 2020 - Apr 2023

- Implemented Flask services to start and stop Jupyter notebooks and persist files in AWS EFS for Pelion's embedded-ML platform; defined PostgreSQL schemas and built end-to-end tests.
- Built sensor-ingestion and dashboard APIs with InfluxDB, including a fixed-radius neighbor algorithm for workspace proximity detection.

Graduate Software Engineer Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware for I2C, GPIO, ADC, DAC and PWM; contributed to shield/form-factor PCBs. 2020 Arm Quality Award (team).
- Built Node.js/Express REST APIs, PostgreSQL schemas and React dashboards to monitor occupancy from thermal cameras and PIR sensors.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Rust, SQL
Applications: Angular, RxJS, Electron, React, Tauri 2, Flask, Node.js, PostgreSQL
Quality / delivery: Playwright, Karma, Jenkins, Git, LSF; SVG/Pixi visualization

EDUCATION

The University of Texas at Austin
Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits
1st Place (Industry), ECE Capstone Design Award - IoT Wireless Home Security System (2018)